

SYDNEY RICHMOND, Australia

WMO: 947530

Lat: 33.6003S

Lon: 150.7761E

Elev: 67

StdP: 14.66

Time Zone: 10.00 (AUS)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.2	34.5	26.7	20.7	56.7	29.1	23.2	53.0	23.8	62.7	20.9	62.9	0.7	90	0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.5	98.5	70.1	93.5	69.6	89.5	68.8	74.9	87.4	73.4	84.7	72.1	82.6	11.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.8	117.8	78.1	70.2	111.4	76.6	69.5	108.5	75.8	38.5	87.7	37.1	84.6	35.9	81.9	88.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.8	18.0	15.7	DB	27.2	107.0	2.0	4.2	25.7	110.0	24.5	112.5	23.4	114.8	21.9	117.8	(4)
(5)			WB	26.9	79.0	2.0	2.8	25.4	81.1	24.2	82.7	23.1	84.3	21.6	86.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	75.3	73.9	70.6	64.3	57.2	52.8	51.3	53.4	59.3	64.4	68.7	72.4	(6)	
	DBStd	9.71	5.87	5.24	4.55	4.49	4.33	4.29	4.13	4.38	5.06	5.76	6.01	5.66	(7)	
	HDD50	68	0	0	0	0	2	20	32	14	0	0	0	0	(8)	
	HDD65	1767	1	1	7	65	244	365	425	359	183	83	28	6	(9)	
	CDD50	5027	785	670	637	428	226	104	72	120	281	446	562	696	(10)	
	CDD65	1251	321	252	179	43	2	0	0	1	13	64	139	237	(11)	
	CDH74	10830	2765	1897	1155	366	33	1	5	25	317	862	1349	2055	(12)	
	CDH80	4556	1338	796	373	86	1	0	0	1	90	343	598	930	(13)	
(14) Wind	WSAvg	5.7	6.7	6.1	5.5	4.5	4.1	4.5	5.0	5.9	6.4	6.4	6.7	6.8	(14)	
(15) Precipitation	PrecAvg	33.1	3.9	4.0	3.6	2.9	2.2	2.4	1.4	2.0	1.5	2.4	3.4	3.0	(15)	
	PrecMax	57.7	12.7	16.6	9.1	12.5	8.7	10.5	5.1	13.6	6.6	6.6	15.6	9.1	(16)	
	PrecMin	15.6	0.3	0.3	0.2	0.1	0.2	0.1	0.0	0.0	0.1	0.0	0.4	0.1	(17)	
	PrecStd	9.3	3.0	3.7	2.4	3.1	2.0	2.4	1.4	3.0	1.3	1.9	2.9	2.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	105.9	101.8	94.0	88.1	77.6	70.8	73.5	77.1	88.1	94.6	100.2	101.6	(19)	
		MCWB	71.7	72.3	70.3	65.3	60.7	58.1	57.8	56.4	61.6	64.3	67.1	69.6	(20)	
	2%	DB	98.6	94.6	88.0	81.9	73.6	67.2	68.0	73.0	82.4	88.5	93.1	95.1	(21)	
		MCWB	72.1	72.4	70.0	65.0	59.8	56.0	53.8	55.8	59.5	63.7	67.5	69.4	(22)	
	5%	DB	92.8	88.8	84.1	77.8	70.6	64.6	64.6	69.3	77.1	83.4	86.2	89.6	(23)	
		MCWB	71.3	71.5	69.0	64.0	58.6	54.9	53.0	54.4	58.7	62.6	65.9	68.2	(24)	
	10%	DB	87.5	84.3	80.4	74.8	68.0	62.6	62.3	65.9	72.8	78.4	81.2	84.6	(25)	
		MCWB	70.5	70.0	68.0	63.0	57.5	54.0	51.8	52.8	57.3	61.5	65.2	67.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.6	77.5	74.8	69.9	66.2	61.2	60.8	60.8	64.8	69.2	73.2	74.7	(27)	
		MCDB	93.3	92.6	83.7	78.5	69.7	66.4	66.9	71.1	78.9	81.4	86.5	88.7	(28)	
	2%	WB	75.1	74.8	72.6	67.5	62.9	59.1	57.1	58.2	62.7	66.9	70.5	72.6	(29)	
		MCDB	88.7	86.2	81.3	75.4	68.6	63.0	62.8	67.2	74.5	79.0	83.7	86.4	(30)	
	5%	WB	73.5	73.2	71.0	66.1	61.1	57.6	55.1	56.3	60.9	65.1	68.7	70.9	(31)	
		MCDB	85.8	82.5	79.2	73.3	67.3	61.5	61.0	64.8	71.4	76.8	80.8	83.0	(32)	
	10%	WB	72.0	71.8	69.6	64.7	59.4	56.1	53.5	54.6	59.1	63.4	67.1	69.5	(33)	
		MCDB	82.9	80.7	76.9	71.8	65.8	60.2	59.7	62.9	69.7	74.2	77.6	79.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	21.5	19.4	19.0	21.1	23.1	21.4	23.9	25.8	25.6	24.2	21.9	21.8	(35)	
		MCDBR	32.8	28.7	26.0	26.7	26.8	24.1	28.1	31.5	34.6	34.8	33.1	32.2	(36)	
	5% WB	MCWBR	10.3	9.6	10.1	12.0	14.5	14.6	16.2	16.6	15.3	13.7	11.9	10.6	(37)	
		MCWBR	26.2	22.9	21.3	21.7	21.5	18.4	21.7	25.8	27.9	26.6	26.2	25.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.397	0.393	0.363	0.333	0.316	0.307	0.298	0.305	0.337	0.361	0.389	0.395	(40)	
		taud	2.430	2.464	2.529	2.585	2.580	2.607	2.614	2.585	2.492	2.453	2.417	2.416	(41)	
	Ebn at Noon		298	293	291	283	272	266	276	289	294	299	299	300	(42)	
		Edh at Noon	39	36	32	27	24	22	23	26	32	36	39	40	(43)	
(44) All-Sky Solar Radiation	RadAvg		2028	1734	1486	1213	971	769	901	1186	1571	1804	1904	2087	(44)	
	RadStd		158	165	137	106	69	67	67	73	93	180	182	202	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+1.13	N/A	N/A	N/A	+371	+279
(47) Regional (37 neighbors)		+0.79	N/A	N/A	+1.39	N/A	N/A	-4	-155	+333	+178

Nomenclature: See separate page